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Students' Thinking About Effort and Ability: The Role of Developmental, Contextual, and Individual Difference Factors

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Students' thinking about the relation between effort and ability can influence their motivation, affect, and academic achievement. Students sometimes think of effort as inversely related to ability (such that people with low ability must work harder than people with high ability) and other times think of effort as positively related to ability (such that hard work can lead people to develop high levels of ability). The purposes of the present review are (a) to review literature on developmental, contextual, and individual difference factors that influence students' thinking about the relation between effort and ability in school and (b) to identify unresolved questions in this literature and present an extended theoretical framework that can help answer these questions. By providing researchers with a better understanding of how students think about effort and ability, we hope that this review will inspire new research in this area.

Keywords: effort, ability, beliefs, attributions

Increasingly, educators and policymakers are turning their attention to "non-cognitive" factors, such as motivation, in order to explain why some students (and groups of students) are more successful academically than others despite receiving the same instruction. One of the strongest predictors of students' motivation and achievement is their self-assessments of ability in different domains (e.g., Schunk & DiBenedetto, 2016; Wigfield, Tonks, & Klauda, 2016). The more competent or efficacious students feel in a particular domain, the more they value it, the higher they set their goals and aspirations, the more frequently they use self-regulatory strategies, the longer they persist on challenging tasks, the less negatively they respond to failure, and the better they perform on academic assessments (e.g., Bandura & Schunk, 1981; Bouffard-Bouchard, Parent, & Larivee, 1991; Denissen, Zarrett, & Eccles, 2007; Durik, Vida, & Eccles, 2006; Hu, Chen, &

Tian, 2016, Pajares, 1996; Simpkins, Davis-Kean, & Eccles, 2006; Trautwein, Lüdtke, Roberts, Schnyder, & Niggli, 2009; Wang, 2012).

When evaluating their own abilities, students draw on many types of information. For example, a student may form an assessment of her own math ability by accounting for recent performance on math tests and assignments (e.g., Marsh & Craven, 2006), as well as what her parents, teachers, or peers have to say about her math ability (e.g., Bandura, 1997). In addition to these external sources of information, students draw on internal cues. In particular, they are attuned to how much effort they have expended on various tasks in a domain. However, studies by Nicholls (1984), Dweck (1999), and others suggest that the inferences students draw based on their perceived effort depend on whether they are currently thinking in terms of an *inverse* or *positive* relation between levels of effort and ability. When a student thinks in terms of an inverse relation, she views effort as an indication of how hard she *had* to work to complete the task, such that high levels of effort are indicative of relatively low levels of ability. In contrast, a student thinking in terms of a positive relation is more likely to view effort as something necessary for improving her overall level of competence in that domain, such that high levels of effort are interpreted as a sign of high levels of ability (e.g., Dweck, 1999; Nicholls, 1984).

In addition to explaining how they form judgments of ability based on their perceptions of past effort, students' thinking about the *relation* between effort and ability may directly influence how hard they decide to work on future tasks (e.g., Blackwell, Trzesniewski, & Dweck, 2007; Dweck & Master, 2009). There is considerably less research on this process; however, it seems that, even when students perceive themselves as having similar levels of ability, those who think of ability as a capacity that dictates how much effort they need to put into a task (i.e., who are thinking in terms of an inverse relation) may exert less effort on an upcoming task than students who think of ability as a set of skills that can be improved over time (i.e., who are thinking in terms of a positive relation). This seems like it may be particularly true when the students are unconfident about their ability to perform well on the task. In this case, a student thinking in terms of an inverse relation may conclude that working harder than other, more talented students (but still performing poorly) will only demonstrate to everyone how little ability she actually possesses; and this may lead her to decide that it is not worth working hard on the task. In contrast, a student thinking in terms of a positive relation may feel that a large amount of effort is worthwhile, even if does not result in good performance, because it will still lead to lasting *increases* in her mastery.

Considering the important influence that students' thinking about effort and ability can have on their motivation and achievement, it is important to understand why they sometimes think in terms of an inverse relation and other times think in terms of a positive relation. With such an understanding, educators will be better equipped to help students focus on the positive relation between effort and ability (which appears to be constructive in most contexts). However, as we discuss later in the article, there may also be contexts in which educators want students to acknowledge the inverse relation that exists between effort and ability in the short term, without losing sight of the long-term positive association between these constructs.

In the present article, we review three types of factors that help explain differences in students' thinking about the relation between effort and ability: students' cognitive maturity, the academic situation or context they are in, and their personal beliefs about the nature of intellectual ability. Although previous reviews have been published that examine students' conceptions of ability (e.g., Bong & Skaalvik, 2003; Butler, 2005; Holloway, 1988; Kinlaw & Kurtz-Costes, 2003; Li & Lee, 2004; Rosenholtz & Simpson, 1984), only Surber (1984b) primarily focused on students' thinking about the *relation* between levels of effort and ability, and this review was published over 30 years ago.

In addition to reviewing research on three types of factors that influence this relation, we point out some inconsistencies and complexities that emerge from a comparison of the somewhat separate (but overlapping) literatures on these factors. We then attempt to make sense of these complexities by presenting an extension of an existing framework by Nicholls (1984). In the original framework, the way that students think about of the relation between effort and ability depends in part on how they currently conceptualize ability itself: If they conceive of ability as an underlying capacity, they are likely to think in terms of an inverse relation between effort and ability, but if they conceive of ability as a set of skills or competencies, they are more likely to think in terms of a positive relation.

As an extension of this framework, we attempt to link these two conceptions into a single understanding of ability that we believe nearly all students possess. According to this understanding, capacity is the *rate* at which skills can be acquired or performed (i.e., the amount of time and effort it takes an individual to complete a task requiring a particular amount of skill). Thus, students who think of ability as a capacity tend to assess their level of ability in a particular domain in terms of how quickly or effortlessly they can complete a task in that domain relative to others (i.e., in terms of an inverse relation between effort and ability). But, as we explain in more detail later in the article, the same students may still acknowledge that this rate can be increased over longer periods of time through consistent hard work and thus, in certain cases, assess their ability in terms of a positive relation.

Furthermore, whether a student who is focused on capacity accounts for the long-term positive relation between effort and ability may depend on whether she believes that ability/intelligence is malleable (i.e., whether she has a fixed or growth mind-set). This last point helps us integrate Dweck's (1999) research on individual differences in students' thinking about effort and ability with research on situational and developmental differences. Hopefully, this updated framework can mitigate some of the confusion concerning these different types of factors and set the stage for future studies that identify and explore novel influences on students' thinking about effort and ability. This work may help educators better understand how students think about effort and ability in school contexts, including why some students are more engaged in academic tasks than others despite having similar levels of ability. Ultimately, this work may also lead to interventions that promote beliefs about effort and ability that are supportive of both short- and long-term academic engagement and achievement.

The review is organized as follows: First, the process of choosing articles for the literature review is described. Second, we provide historical context by briefly

reviewing attribution theory. Third, we review the literature on the developmental, contextual, and individual difference factors that influence students' thinking about effort and ability. Fourth, we point out some methodological differences and empirical inconsistencies within this literature and present an extended theoretical framework for understanding students' thinking about effort and ability. Fifth, we will discuss new developments in the literature, specifically recent work on perceived effort source (Muenks, Miele, & Wigfield, 2016). Finally, we will use the current literature and our extended framework to provide suggestions for future research or interventions aimed at increasing students' motivation and achievement in school.

Method

This literature review is primarily based on searches for journal articles using Academic Search Complete in EBSCOHost. There were four target words or categories of words: (a) words about ability ("ability," "intelligence," "gifted," or "smartness"); (b) "effort"; (c) words about beliefs ("beliefs," "theories," or "conceptions"); and (d) "academic." We searched for articles published in peer-reviewed journals between January 1970 and February 2016 that were published in English and were conducted in the United States. This search produced a total of 435 articles. After reading the abstracts of each of these articles, we narrowed down our search further by only including studies that met the following criteria: (a) studies that examined the *relation* between conceptions of effort and ability; (b) studies that examined effort and ability in academic settings; (c) studies that utilized samples of participants aged 5 years and older who were not diagnosed with learning disabilities; and (d) studies that were quantitative in nature (e.g., experimental, quasi-experimental, correlational, longitudinal, cross-sectional, or intervention).

After narrowing down our search based on these criteria, we were left with a total of 86 articles. The first author read all 86 articles and made decisions about which of these articles to include in the review based on the scientific rigor of the studies and their relevance. In addition, we conducted supplementary searches of reference lists and also used Google Scholar and Web of Science to identify additional articles that were relevant to this review but that may have been overlooked in the initial search. Some of these articles included samples of participants from outside of the United States.

Results

Historical Background: Attribution Theory

In order to better understand current research on the perceived relation between levels of effort and ability, it is useful to consider the ways in which these concepts were originally articulated as part of attribution theories of motivation. Attribution theorists such as Heider (1958) and Weiner (1985) originally aimed to explain how individuals interpret outcomes and events in terms of their causes. Educational psychologists, including Weiner himself, have applied these theories to make sense of the ways in which students are motivated to respond to academic successes and failures.

Heider (1958) originally distinguished between two different conditions of action: "can" and "try." "Can" was defined as "the relation between the power or ability of the person and the strength of the environmental forces" (Heider, 1958, p. 86), such that only those with high levels of ability (defined as a stable, dispositional characteristic) can succeed at tasks that are particularly difficult or challenging. "Try," which is similar to effort, also includes two elements: intention and exertion. Intention is defined as wishing or wanting to do something, whereas exertion is how hard a person tries to do it. Heider described exertion as varying directly with the difficulty of the task and inversely with the ability of the person. In other words, if task difficulty is held constant, a person with low ability will have to exert herself more than a person with high ability in order to complete the task.

Weiner (1986), whose theory has been particularly influential within educational psychology, also focused on effort and ability as central causes of success or failure at academic tasks (along with task difficulty and luck). He delineated three dimensions along which attributions can be categorized: locus, stability, and controllability. The most fundamental dimension is locus, which describes whether the cause is perceived as internal or external to the individual. Ability and effort are both categorized as internal factors. The second dimension is stability. Although in some of his earlier work, Weiner (1979) distinguished between *stable* effort that reflects individual differences in how much effort people generally expend and *immediate* effort that emerges from the current situation. In his later work, he focused primarily on immediate effort and characterized it as variable and unstable. He characterized ability, on the other hand, as a stable and constant capacity.

The third dimension is controllability, which is defined as whether the individual has volitional control over the perceived cause. Effort is described as controllable, whereas ability is uncontrollable. Thus, Weiner (1986) characterized ability as internal, stable, and uncontrollable, and effort as internal, unstable, and controllable. These characterizations of ability and effort suggest that, in achievement situations, individuals are likely to perceive them as inversely related: When ability is stable and uncontrollable, it cannot be substantially increased with effort. In sum, attribution theorists introduced the idea that students' attributions of success and failure to effort and ability are motivationally important and set the stage for researchers such as Nicholls (1978, 1984) and Dweck (1999) to more directly examine how students perceive the *relation* between effort and ability, which is the focus of this review paper.

Developmental Influences on Students' Thinking About Effort and Ability

How young children conceptualize the relation between effort and ability depends, in part, on their understanding of the concepts of effort and ability. John Nicholls's (1978) developmental theory of ability conceptions builds on attribution theory by specifically examining what developmental factors influence how individuals think about effort and ability in school-related contexts. In Nicholls's (1978) original study, participants aged 5 through 13 years were shown videos of two 8-year-old children working on math problems at their desks. One child in the video appeared to be working hard, whereas the other child did not. Participants were then told either that both children got the same number of answers correct on

the math problems, or that the child who worked harder got a lower score than the other child. The participants were then asked to determine which child from the video was smarter and explain their reasoning.

Based on a qualitative analysis of the data, Nicholls (1978) proposed four levels of reasoning about others' effort and ability. In the first level (age 5–6 years), effort, ability, and outcome are not distinguished from each other. Children at this age tend to focus primarily on effort as an indicator of high ability. For example, they believe that the character from the video that tried harder is smarter, even if he or she received the lower score. In the second level (age 7–9 years), effort and outcome are distinguished as cause and effect (i.e., different levels of effort lead to different outcomes), but ability is not acknowledged as a cause of outcome because it is still conflated with effort. In the third level (age 9–11 years), the distinction between effort and ability is partially established; effort is no longer the sole cause of outcomes. However, this reasoning is not consistently applied. In the fourth level (age 10–13 years), ability is clearly differentiated from effort, and the two are seen as independent causes of outcomes. Furthermore, ability is seen as a capacity that can limit or increase the effectiveness of effort. Nicholls (1978) suggested that most children go through this series of developmental levels in their reasoning about effort and ability and eventually come to the "mature" conclusion that levels of effort and ability are inversely related when outcome is controlled (e.g., You need less of one if you have more of the other to reach the same outcome).

Two central claims of Nicholls's (1978) developmental theory can be abstracted. First, young children do not conceptually differentiate between effort and ability, but older children and adults do. And, second, as children develop, they become capable of thinking about effort and ability as being inversely related. Both of these claims have received support from numerous empirical studies. To start with, several studies have provided support for the idea that young children cannot differentiate effort from ability (e.g., Bempechat, London, & Dweck, 1991; Roque, de Lemos, & Gonçalves, 2014; Skinner, 1990; Stipek & Daniels, 1990; Yussen & Kane, 1985). For example, Stipek and Tannatt (1984) interviewed children in preschool through third grade about other students in their class and found that for children of all ages, the concepts of effort, ability, and conduct were confounded.

With regard to Nicholls's (1978) second claim, Kun (1977) explored two inference rules that people use when thinking about effort and ability: first, a magnitude-covariation schema, which involves understanding that an increase in the magnitude of an effect (e.g., success on a task) is preceded by an increase in the magnitude of a cause (e.g., ability or effort); and, second, a compensation schema, which involves understanding that when an effect is invariant (e.g., when performance neither improves nor declines), a change in the magnitude of one cause is accompanied by a compensating change in the magnitude of a second cause. Kun (1977) distinguished between two types of compensation schemas: A direct compensation schema, where the strength of the two causes changes in the *same* direction (e.g., an increase in effort occurs at the same time as an increase in task difficulty), and an inverse compensation schema, where the strength of the two causes changes in *opposite* directions (e.g., an increase in ability is counteracted

by a decrease in effort). Utilizing samples of elementary school students and undergraduates, Kun (1977) found that the magnitude covariation schema develops first, then the direct compensation schema, and finally the inverse compensation schema. Together, these results suggest that as children get older, they learn to combine effort, ability, and outcome information in ways that are consistent with an inverse relation between effort and ability.

A number of other researchers have explored similar developmental changes in children's reasoning about effort and ability (e.g., Droege & Stipek, 1993; Graham & Barker, 1990; Juvonen & Murdock, 1995; Miller, 1985; Nicholls & Miller, 1984). For example, Nicholls and Miller (1984) extended Nicholls's (1978) original study by including self-evaluations of effort and ability in addition to evaluations of others. Results supported Nicholls's (1978) claim that, by Grades 5 or 6, children perceive an inverse relation between levels of effort and ability (in self and others). Additionally, Folmer et al. (2008) replicated Nicholls's original findings (Nicholls, 1978; Nicholls & Miller, 1984) and extended them to situations involving *negative* performance outcomes. Overall, the findings from these studies suggest that, as children get older, they come to understand that effort and ability can be inversely related (see also Freedman-Doan et al., 2000; Kurtz-Costes, McCall, Kinlaw, Wiesen, & Joyner, 2005; Rholes, Blackwell, Jordan, & Walters, 1980; Rholes, Jones, & Wade, 1988; Rholes & Ruble, 1984).

Several additional studies provided support for the idea that the perceived relation between effort and ability changes over time but did not provide support for the specific age-related levels proposed by Nicholls (1978). For example, several researchers have demonstrated that children as young as 4 to 6 years show an understanding of the inverse relation between effort and ability (e.g., Heyman & Compton, 2006; Heyman, Gee, & Giles, 2003; Karabenick & Heller, 1976; Schunk, 1983; Surber, 1980; Wimmer, Wachter, & Perner, 1982). Conversely, Harari and Covington (1981) found that a fully mature understanding of the relation between effort and ability did not occur until 13 to 14 years of age, which is later than Nicholls's (1978) theory predicted.

Why is there such disagreement in the literature about when children are able to understand the inverse relation between effort and ability? There are a number of potential explanations. Several of the researchers that found that children were able to understand the inverse relation between effort and ability prior to Nicholls's (1978) original findings followed specific precautions to ensure that the younger children fully understood the task. For example, Karabenick and Heller (1976) provided information verbally to participants in a cascading pattern. They first provided outcome information (e.g., "Tom successfully completed a puzzle") and had children make an ability (or effort) evaluation. They then provided effort (or ability) information (e.g., "Tom worked very hard") and allowed the children to change their initial ability (or effort) evaluation. This way of organizing the task may have helped young children consider the unique contribution of each piece of effort or ability information (see also Surber, 1980; Wimmer et al., 1982).

Perhaps when researchers reduced the cognitive demands of the task, younger children were able to show evidence of relational thinking about effort and ability and performed similarly to the older children in other studies (see also Singh & Singh, 1994). Indeed, Surber (1984b) suggested that reasoning in terms of an

inverse relation between effort and ability is more cognitively demanding than reasoning in terms of a positive relation and, thus, may not emerge for younger children in studies where processing demands are high. In addition to the cognitive demands of the tasks, there are several other factors that could have contributed to differences between studies, including domain (e.g., academic vs. nonacademic), materials and measures (e.g., text-based vignettes, video recordings, or more "real-life" scenarios in which participants actually completed tasks), information about task difficulty (see Surber, 1984b), and sample characteristics (the samples tended to be fairly small overall and were recruited from different countries around the world).

Conclusion

In summary, the research described above supports Nicholls's (1978) claim that most individuals eventually come to understand that levels of effort and ability can be inversely related to one another. However, it is possible that certain developments in children's reasoning about effort and ability may occur earlier or later than Nicholls (1978) predicted. Although the researchers described above did not all agree on the specific ages at which that these conceptions arise, they did agree that, by adulthood, most people understand and can fairly consistently use the inverse compensation rule when evaluating ability level based on information about effort level.

Contextual Influences on Students' Thinking About Effort and Ability

Importantly, just because mature individuals are *able* to view levels of effort and ability as inversely related does not necessarily mean that they will do so in every context. As an extension of his developmental theory, Nicholls (1984) suggested that students' thinking about the relation between ability and effort depends on their current conception of ability, which is influenced by specific features of achievement situations. In situations where individuals are seeking to demonstrate or assess normative levels of ability (i.e., *ego-involved* situations) or where there is an emphasis on social comparison and competition, individuals tend to employ what Nicholls (1984) calls a "differentiated" conception of ability, such that ability is seen as an underlying capacity that constrains the effects of effort on skill development or knowledge acquisition. In these situations, individuals view high effort as an indication that this capacity is lacking; that is, the more effort or time it takes to learn something, the lower one's capacity. Thus, in ego-involved contexts, individuals are likely to perceive an inverse relation between levels of effort and ability. However, in situations where people are focused on increasing their mastery or competence (i.e., *task-involved* situations) or where there is less of an emphasis on social comparison, individuals are more likely to adopt an "undifferentiated" conception of ability, such that ability is equated with the skills or knowledge that one develops through effort. In these situations, individuals believe that more effort will lead to greater learning and, thus, perceive a positive relation between levels of effort and ability.

In a series of studies, Jagacinski and Nicholls (1984) asked college students to imagine themselves in a hypothetical task-involved or ego-involved scenario where they put forth either high or low effort and to then rate their level of

competence. They found that in the ego-involving conditions (e.g., studying Italian to become an Italian professor, a job with a scarcity of positions available), less effort was associated with higher competence ratings, whereas in task-involving conditions (e.g., studying Italian due to a love of Italian culture and interest in the language), more effort was associated with higher competence ratings.

In another series of studies, Jagacinski and Nicholls (1987) sought to disentangle the influence of social comparison information from task/ego involvement on individuals' conceptions of ability. They asked college students to imagine themselves in either a task-involved or ego-involved situation and then either presented or withheld social comparison information. They found that, in the absence of social comparison information, college students who were asked to imagine that they had exerted high levels of effort in order to master a task (who were in the task-involving condition) reported greater feelings of competence than students who imagined that they had exerted low levels of effort, consistent with the results of the prior study. However, once social comparison information was emphasized (e.g., when students were told, "Other people did not find [the task] challenging and needed much less effort than you did"), even the students in the task-involving condition associated higher effort on the task with *less* competence. Overall, these studies show that aspects of the situation (i.e., whether task-involved, ego-involved, or emphasizing or deemphasizing social comparison information) can influence one's perception of the relation between levels of effort and ability.[1]

Conclusion

In summary, this situational component of Nicholls's (1984) theory extends his developmental component by explaining why older children and adults, once they are able to conceptualize ability as either differentiated or undifferentiated from effort, appear to alternate between these conceptions of ability. In ego-involved situations, where social comparison and competition are salient, more effort is indicative of lower levels of "differentiated" ability (an inverse relation); but, in task-involved situations, where the focus is on individual growth, more effort is a sign of higher levels of "undifferentiated" ability (a positive relation). Thus, situational cues are important determinants of how individuals think about effort and ability within a particular context.

However, the few studies that have demonstrated this are somewhat dated and researchers utilized college student samples in tightly controlled environments (e.g., Jagacinski & Nicholls, 1984, 1987). No researchers that we know of have directly explored the effects of contextual cues on the perceived relation of effort and ability in actual classrooms. Future researchers who attempt to address this gap in the literature could take advantage of sophisticated data analytic approaches, such as hierarchical linear modeling (Raudenbush & Bryk, 2002), which would allow researchers to investigate classroom effects while controlling for individual differences in students' thinking about effort and ability within each classroom. We review the literature on these individual differences in the next section.

Individual Difference Influences on Students' Thinking About Effort and Ability

In addition to developmental and situational factors, there are also stable individual differences in people's conceptions of ability that affect the way they view

the relation between levels of effort and ability *within* particular contexts. In her review of the literature on how individuals think about relations between effort and ability, Surber (1984b) discussed several studies that pointed to individual differences in children's and adults' use of the inverse compensation rule. For example, she reanalyzed the results of Surber (1980) and found that, for each of the elementary school age groups, some participants judged ability to be positively related to effort, whereas others judged ability to be inversely related to effort. When looking at the sample as a whole, these patterns were hidden. These same individual differences were found in Kun's (1977) data. In Karabenick and Heller (1976), only about 60% of the elementary school participants used inverse compensation on both trials of the experiment, suggesting that a substantial number of students used a different rule at least some of the time. Surber (1984a) asked college students to evaluate the ability of hypothetical students when provided study effort and performance information and found that students could be categorized into two different groups. Approximately half the sample viewed effort and ability as positively related, whereas the other half viewed effort and ability as inversely related.

Surber (1984b) provided two different potential explanations for these individual differences. First, individuals with lower cognitive capacity may have relied on more "immature" processing strategies when completing these tasks and thus viewed a positive relation between effort and ability, whereas individuals with higher cognitive capacity may have thought in terms of an inverse relation, which is more cognitively challenging (this is similar to her account of developmental differences, see above). Second, it may have to do with how individuals conceptualize ability. In particular, Carol Dweck and colleagues have conducted a great deal of research on people's lay theories about the fixedness of intellectual ability and intelligence (see Dweck & Molden, 2005).[2]

Dweck (1999; Dweck & Master, 2009) posited that individuals who believe that intellectual ability is fixed and unchanging (i.e., entity theorists) tend to believe that no amount of effort will change one's ability, and that, in fact, expending a lot of effort is an indication that one's innate level of ability is more constrained or limited than the ability of an individual who did not have to put in as much effort in order to be successful on the same task. Thus, these individuals are likely to perceive amount of effort as *inversely* related to ability across contexts. Conversely, those who believe that intellectual ability is malleable (i.e., incremental theorists) tend to believe that increased effort actually leads to improvement and mastery. Thus, they are more likely to perceive amount of effort as *positively* related to ability across contexts.

Evidence for these hypotheses comes from a study by Hong, Chiu, Dweck, Lin, and Wan (1999) in which college students were induced to temporarily adopt either an entity or incremental theory of intelligence and were later presented with a hypothetical scenario in which they were asked which of two students was more intelligent: Student A, who ranked 1st in the class and was diligent, or Student B, who ranked 10th in the class but only studied before the test. The participants induced to hold an entity theory were more likely than the participants induced to hold an incremental theory to name Student B as the more intelligent student,

suggesting that an entity theory can lead student to perceive low levels of effort as indicative of high levels of intellectual ability. This study is of particular importance because it demonstrates (a) that people's implicit theories of intelligence, though stable, can be changed and (b) that these theories have a causal effect on people's thinking about the relation between levels of ability and effort.

In a number of other studies, researchers found additional support for these conclusions. In one study, Mueller and Dweck (1998; as cited in Dweck, 1999) found that college students who held an entity theory (identified using a median split) were likely to agree with statements such as, "I sometimes feel that the more effort you have to put into your school assignments, the less intelligent you probably are," whereas incremental theorists were more likely to disagree with these statements. In another study, Dweck (1999) found that grade school children with an entity theory of ability said they felt smart "When I turn in my papers first," whereas children with an incremental theory said they felt smart "When I don't know how to do it and it's pretty hard and I figure it out without anybody telling me."

Miele and colleagues (Miele & Molden, 2010; Miele, Son, & Metcalfe, 2013) found that entity theorists in both elementary school and college were less confident about their comprehension when they expended relatively high levels of effort reading a text, whereas this was not the case for incremental theorists. They interpreted these findings as evidence that the entity theorists construed their own experiences of mental effort as an indication that they were reaching the limits of their ability, whereas incremental theorists were more likely to construe effort as a sign that they had overcome challenge or increased their mastery. In a longitudinal study, Blackwell et al. (2007) found that an incremental theory was associated with positive effort beliefs (e.g., "the harder you work at something, the better you will be at it") and that these beliefs partly mediated the positive effect of holding an incremental theory on students' math grades (see also Jones, Wilkins, Long, & Wang, 2012; Rickert, Meras, & Witkow, 2014; Tempelaar, Rienties, Giesbers, & Gijselaers, 2015).

Conclusion

As a whole, the studies reviewed above provide support for Dweck's (1999; Dweck & Master, 2009) theory concerning individual differences in people's beliefs about intellectual ability and can help account for students' perceptions of both positive and inverse relations between levels of effort and ability. Individuals who believe that ability is fixed and unchanging tend to view effort and ability as being inversely related in particular situations because they conceptualize ability as an innate capacity, whereas those who believe that ability is malleable tend to view effort and ability as positively related because they conceptualize ability as consisting of skills or competencies that can change over time.

The literature reviewed here includes both correlational and experimental designs, and the samples span a broad range of ages. However, there are some inconsistencies between this work and the research described in the developmental and contextual sections above. Next, we point out some of these inconsistencies and discuss ways in which they may be resolved.

Unresolved Questions

Over the past 45 years, researchers have found convincing evidence that developmental, situational, and individual difference factors influence whether individuals think in terms of a positive or inverse relation between effort and ability. However, there are some unresolved questions concerning the circumstances that lead certain factors to take precedence over others. For example, in some studies, social comparison or ego involvement cues (such as asking participants to compare the abilities of two students) appear to lead almost *all* cognitively mature individuals to view effort and ability as being inversely related. For example, Karabenick and Heller (1976) found that 91% of college students made ability attributions and 88% made effort attributions consistent with an inverse relation between effort and ability, and Nicholls et al. (1986) found that 100% of college students exhibited Level 4 reasoning about effort and ability (i.e., which involves viewing low effort as indicative of high ability, when performance is held constant). Thus, presumably, regardless of any individual differences in students' theories of intelligence, almost all of the participants across the two studies endorsed an inverse relation between effort and ability. However, in other studies, individual differences in students' theories of intelligence do affect how they view the relation between effort and ability, despite being situated in ego-involved contexts (e.g., Blackwell et al., 2007; Hong et al., 1999). This raises the question of what circumstances lead to situational cues taking precedence over individual difference factors and vice versa.

One possible answer pertains to methodological differences between the studies discussed in the sections on developmental and contextual factors and those discussed in the section on individual differences. In developmental studies where participants were asked to compare the abilities of two students who had put forth different amounts of effort (Folmer et al., 2008; Nicholls, 1978; Nicholls et al., 1986), these students were always described as having exhibited the *same* level of performance. In Hong et al.'s (1999) study, however, the student who put forth more effort was described as having *higher* grades than the student who put forth less effort. What this suggests is that certain situational cues may allow for more direct comparison between students, which can lead to a differentiated conception of ability. That is, when the performance of two students is equated, participants are able to make inferences about which student needed less effort to achieve a certain performance outcome and, thus, tend to think in terms of differentiated conception of ability, which in turn leads them to perceive an inverse relation between levels of effort and ability. But, when the situation is more ambiguous because the harder working student is described as having performed slightly better than the other student, some participants may wonder whether the less hard-working student would have performed better *or only the same* as the first student had she worked just as hard. When facing this type of ambiguity, participants may be more likely to rely on their general (i.e., situation-independent) beliefs about ability, which could explain why many of the incremental theorists in the Hong et al. (1999) study exhibited a positive relation between effort and ability.

Although differences in situational cues (e.g., cues that allow for direct social comparison vs. cues that do not) may partly explain variability in students'

reasoning about effort and ability across studies, they do not explain why or how an individual's subjective definition or conception of intellectual ability can be both stable and variable at the same time. How, for instance, can a student with a stable incremental theory of intelligence (which would seem to be consistent with an undifferentiated conception of ability) suddenly shift to a differentiated conception (e.g., when entering a competitive math classroom) and then back to an undifferentiated conception (e.g., when heading to her more mastery-oriented English class)?

Intuitively, we tend to think of individuals' conceptions as changing somewhat gradually in a particular direction (such as the developmental changes in conceptions of ability described by Nicholls), as opposed to rapidly changing back and forth between contexts (such as the situational changes in conceptions of ability described by Nicholls). When students shift their thinking about the relation between effort and ability across contexts, is this because they are actually defining ability in an entirely different way (i.e., a way that is incompatible with how they thought about ability a few minutes ago), or is it because they are momentarily focused on a different aspect of a more complex conception of ability? In addressing this issue, we present an extension of Nicholls's (1984) original framework that attempts to clarify how students can simultaneously hold different conceptions of ability (see Figures 1a and 1b), and that accounts for Dweck's (1999) work on individual differences in beliefs about ability (see Figure 2).

Extended Theoretical Framework

To begin with, we think it is important to note that the direction of causality may be different when thinking in terms of an inverse or positive relation between effort and ability. For instance, when students think of effort as being inversely related to ability, they are assuming a causal relation from ability to effort; that is, low ability necessitates high levels of effort on challenging tasks, but high effort does not in any way reduce one's level of ability. However, when students think of effort as positively associated with ability, they are more likely to have assumed a causal relation from effort to ability (i.e., by putting in high levels of effort, people are able to increase their ability); though it is still possible they have assumed the opposite relation (e.g., people with high ability are more interested in working hard on academic tasks).

These two causal relations are not mutually exclusive. That is, it is possible for the same individual to think in terms of both a positive relation from ability to effort and an inverse relation from effort to ability within in a given context. For instance, students can believe that (at this point) they have to work harder than their peers in order to complete a challenging task because they currently have a low level of ability, while also believing that their efforts will serve to increase their ability in the end. This framework is depicted in Figures 1a and 1b.

In both figures, units of skill/competence are on the *y*-axis and units of effort are on the *x*-axis. The three lines indicate individuals who begin a particular task with different learning capacities (where capacity is defined as the rate at which one acquires or performs new skills/knowledge/competencies; indicated by the slope of each line). What a comparison of the boxes in Figure 1a suggests is that, if task completion requires 10 units of skill/competence, an individual who begins the task with a high level of ability will need less effort to complete the task (e.g.,

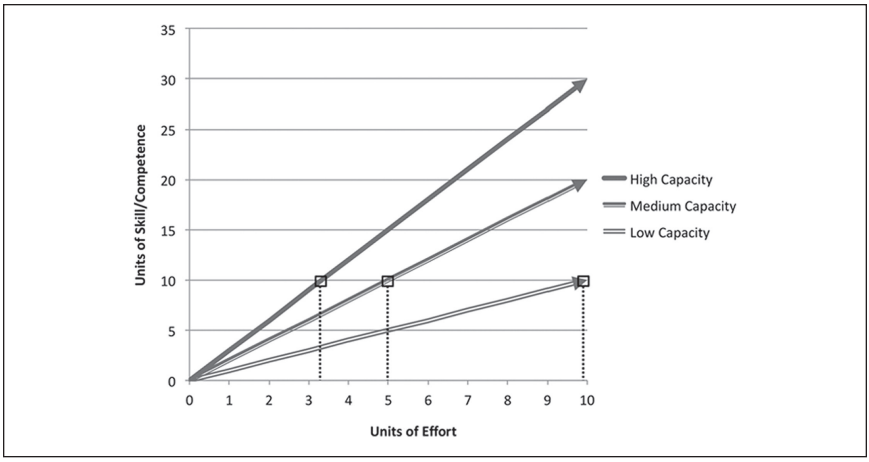

FIGURE 1A. *A focus on ability as capacity (i.e., rate of skill acquisition/performance), leading to an inverse causal relation from ability to effort. Each line represents an individual with a different capacity but with the same initial level of competence. The boxes indicate how much effort each individual will need to complete a task requiring 10 units of competence.*

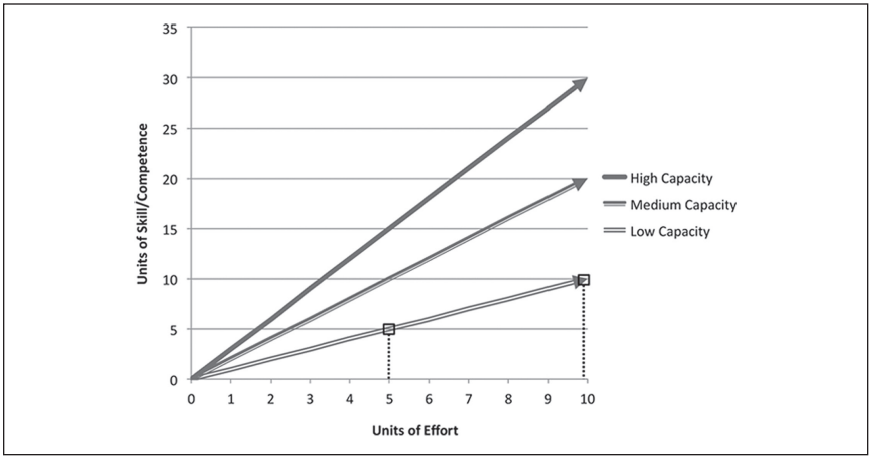

FIGURE 1B. *A focus on ability as overall level of competence, leading to a positive causal relation from effort to ability. Each line represents an individual with a different capacity but with the same initial level of competence. The boxes indicate how much effort the low-capacity individual expends to increase her competence from 5 units to 10 units.*

just over 3 units) than an individual who begins with a low level of ability (e.g., 10 units). By focusing on the aspect of ability that pertains to the *rate* of learning (i.e., capacity), we assume an *inverse* causal relation from ability to effort. In this

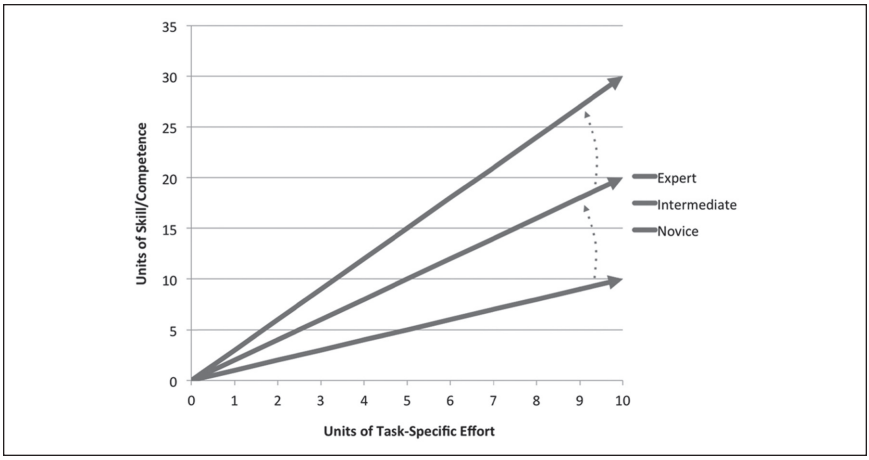

FIGURE 2. *Incremental theorists' hypothetical view of capacity as stable in the short term but malleable in the long term. Each line represents a different level of expertise that can be achieved by the same individual (with experts having a greater capacity in the domain than novices). For a short-term task, the units of effort required to achieve a certain level of competence are determined by the person's current capacity (i.e., ability determines effort). However, as the person gains expertise over longer periods of time (represented by the dotted arrows), her capacity increases (represented by the steeper slope of the top line), such that she will need less effort to achieve the same level of competence on the task as before.*

case, if we know that an individual's effort on the task was high, we are likely to infer that she *had* to work that hard because of her low ability. However, within the same scenario, we can instead focus on the aspect of ability that pertains to what one can actually do (i.e., overall level of competence; see Figure 1b). From this perspective, all three individuals began the task with relatively low levels of ability and, after putting forth a certain amount of effort, ended with higher levels of ability (as should be clear from a comparison of the circles in Figure 1b). Thus, by thinking of ability as set of skills, instead of as a capacity for learning, we come to assume a positive causal relation from effort to ability, such that high levels of effort will eventually lead to the development of high ability.

We can use this framework to better understand the findings of both Nicholls (1978, 1984) and Dweck (1999) reviewed above. Rather than providing a radically different way of thinking about these constructs, we see our framework as elucidating a distinction originally made by Nicholls (1984), and bridging his work with Dweck's (1999). Indeed, Nicholls's (1984) distinction between differentiated and undifferentiated conceptions of ability is analogous to our distinction between the capacity and competence aspects of ability. The key difference is that we are slightly more specific in how we define the perceived relation between capacity and competence. That is, we define capacity as the *rate* at which one acquires or demonstrates a certain level of competence/skill/knowledge. By integrating both aspects of ability (i.e., capacity and competence) into a single conception, we are in a better position

to understand how students can simultaneously perceive both a positive and inverse relation between effort and ability. For example, students in a calculus class can believe that the hard work they put into completing a difficult problem set serves to increase their math skills, while acknowledging that the rate of this increase may be limited by their current capacity and somewhat slower that rates of their peers.

In addition, our framework can be used to explain possible differences between this conception and individuals' theories about the fixedness of intelligence. With respect to Dweck's (1999; Dweck & Master, 2009) theory, there are multiple ways in which entity and incremental theories of intelligence might map onto to the constructs in our framework. First, holding an entity theory might result in a general tendency to focus on the capacity component of ability, whereas an incremental theory might instead be associated with a tendency to focus on the competence level component. However, it is also possible that the distinction between entity and incremental has more to do with what one believes about the nature of intellectual capacities. More specifically, entity theorists might believe that the *rate* at which people acquire skills/competence is fixed (and akin to something like fluid intelligence), whereas some incremental theorists might instead believe that, although this rate tends to be stable in the short term, it is still malleable and can increase gradually over time as one develops expertise in a particular domain (this view, depicted in Figure 2, is consistent with findings from the literature on expertise; see Chi, Glaser, & Rees, 1982).

In this case, whether incremental theorists perceive an inverse or positive relation between effort and ability may depend in part on their frame of reference for interpreting an individual's effort and evaluating her ability. If incremental theorists are focused on the capacity component of ability (because they find themselves in an ego-involved context) *and* they perceive an individual's effort as unfolding over a relatively short period of time (e.g., during a class period), they may view this effort as the product of the individual's current capacity (which is unlikely to change substantially during this brief interval). This in turn would lead them to interpret relatively high levels of effort as indicating low levels of ability (an inverse relation). However, if incremental theorists are focused on capacity and perceive an individual's effort as unfolding over a relatively long period of time (e.g., during a semester or year of college), they may instead view this effort as leading to a gradual increase in the individual's capacity for acquiring new skills/knowledge. Thus, high levels of effort would be seen as resulting in relatively high levels of ability (a positive relation).

This understanding of the difference between an entity and incremental theory of intelligence provides an alternative explanation for the inconsistent findings from the developmental and individual difference studies reviewed above. In the developmental studies by Nicholls and others (e.g., Folmer et al., 2008; Nicholls, 1978), participants were asked to evaluate the abilities of two students who had put forth different levels of effort on a particular task; whereas, in the individual difference study by Dweck and colleagues (Hong et al., 1999), participants were asked to evaluate the abilities of two students who put different levels of effort into studying for a test over the span of many weeks. Thus, the reason why all of the college students in Nicholls et al.'s (1986) study (including the incremental theorists presumably) identified the harder working student as having less ability

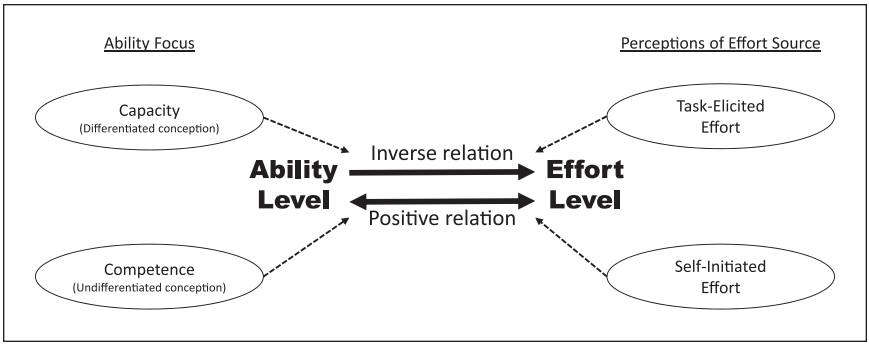

FIGURE 3. *Conceptual diagram depicting how differences in ability focus and perceived effort source can influence students' thinking about the relation between effort and ability.*

may be because they viewed the students' intellectual capacities as being stable over such a short period of time. In contrast, the reason that half of the incremental theorists in the Hong et al. (1999) study identified the harder working student as having more ability may be because they viewed the students' capacities as susceptible to change over relatively long periods of time.

New Developments: Perceptions of Effort

In the studies described earlier, researchers examined how students' conceptions of *ability* (i.e., differentiated vs. undifferentiated, entity vs. incremental), which were influenced by developmental, situational, and individual difference factors, contributed to students' thinking about the relation between effort and ability. However, as illustrated in Figure 3, this only explains half of the equation. That is, differences in the perceived relation between effort and ability may also involve different ways of thinking about the nature of effort. Recent research from educational psychology (Muenks et al., 2016), building on previous findings from the developmental and cognitive literatures (e.g., Heyman & Compton, 2006; Koriat, Ma'ayan, & Nussinson, 2006), has begun to explore this possibility.

Heyman and colleagues (Heyman & Compton, 2006; Heyman et al., 2003) explored how different perceptions of effort can lead to differences in the perceived relation between effort and ability. In Study 1 of Heyman and Compton (2006), 5- to 10-year-old children judged the ability of two different characters in a hypothetical scenario. One of the characters was said to have finished a set of puzzles very quickly, whereas the other character finished the puzzles slowly. This difference in actual puzzle-solving effort was framed differently depending on whether participants were in the *intentional effort* or *perceived difficulty* condition.[3] In the *intentional effort* condition, participants were told that the slow character "tried and tried" and the fast character "hardly tried at all." In the *perceived difficulty* condition, participants were told that the slow character "thought the puzzles were hard to do" and the fast character "thought the puzzles were easy to do." Results showed that the older children in the *perceived difficulty* condition identified the faster child as smarter than in the *intentional effort* condition.

This finding indirectly suggests that information about *why* someone works hard affects how children think about the relation between effort and ability. When the time an individual spent on a task was described in terms of subjective difficulty, children tended to view effort and ability as inversely related, but when time on task was described in terms of goal-directed effort (e.g., "trying hard"), children did not necessarily view effort and ability as inversely related. This difference appeared to be more pronounced for the older children than the younger children.

Muenks et al. (2016) built on the work of Heyman by examining college students' perceptions of the *source* of effort. They also built on work in the cognitive psychology literature by Koriat and colleagues (e.g., Koriat et al., 2006; Koriat, Nussinson, & Ackerman, 2014), who found that students could perceive effort as either a means of *improving* learning and task performance, or as an *indicator* of how much they had learned and how well they will perform. Specifically, Muenks et al. (2016) proposed that when students observe another individual working hard, their interpretation of this effort will depend on whether they believe that it is merely an automatic response to the increasing demands of the task or a deliberate attempt to exceed the minimal level of effort required to complete the task (see Figure 3 for a conceptual model). In the former case, the effort is interpreted as having being elicited by the subjective difficulty of the task and, thus, as a sign of low ability. In other words, the harder an individual *must* work to successfully complete a task in a particular domain, the less ability she must have in that domain; Muenks et al. (2016) refer to this as *task-elicited effort.* In the latter case, the effort is interpreted as resulting from a self-initiated attempt to increase one's mastery or performance and thus does not necessarily signal low ability; Muenks et al. (2016) refer to this *self-initiated effort*. To the extent that self-initiated effort is associated with intellectual growth or high levels of prior achievement (e.g., the idea that smart people are more interested in school), some people may actually interpret it as a sign of high ability.

Muenks et al. (2016) directly tested this proposal by having participants read a series of vignettes and asking them to compare the abilities of two hypothetical characters from each vignette. Participants received information about the level of effort each character exerted, the "source" of their effort (i.e., task-elicited, self-initiated, or control), and their performance on the task. Results indicated that, in the absence of explicit performance information, when effort was described as task-elicited (i.e., when hardworking characters found the tasks to be difficult and characters who did not work hard found the tasks to be easy), participants were more likely to perceive an inverse relation between effort and ability (i.e., to rate the hardworking characters as having less ability than the characters who did not work hard). In contrast, when effort was described as self-initiated (i.e., when hardworking characters were motivated to engage in the task and characters who did not work hard were not motivated), participants were less likely to perceive an inverse relation and, in some cases, were more likely to perceive a positive relation between effort and ability. This pattern of results was also found when participants were asked to recall achievement situations involving peers or acquaintances from their real lives and then evaluate their peers' abilities.

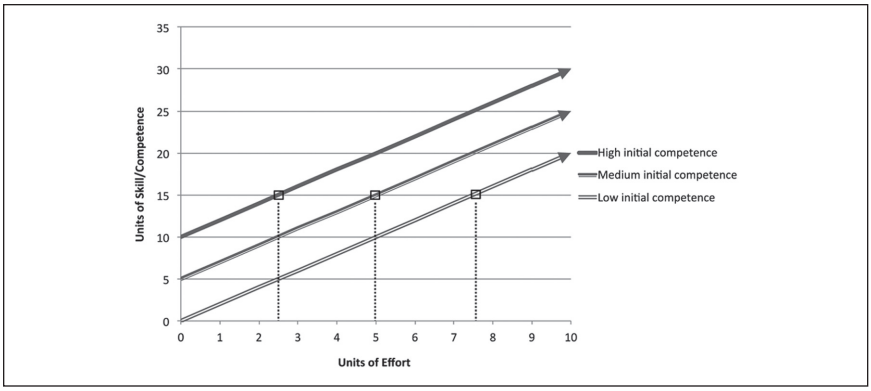

FIGURE 4. *A focus on ability as overall level of competence, when individuals are assumed to have the same underlying capacity (represented by the slope of the lines), but to vary in terms of initial level of competence when starting the task. The boxes indicate that the three individuals must exert different levels of effort to complete a task that requires 15 units of competence, and that by putting in different levels of effort they end up with the same level of competence/ability.*

Findings from the Muenks et al. (2016) studies on perceived effort source can also be understood in terms of the framework we proposed in the previous section (see Figures 1 and 2). Specifically, when students perceive effort to be task-elicited (e.g., Muenks et al., 2016), they are assuming a causal effect of ability on effort, such that an individual must expend a high level of effort *because* she has a low level of initial ability. Whether students conclude that this low level of *initial* ability implies a low level of *current* ability depends in part on whether they assume a competence or capacity view of ability. If they assume a competence view, then they should conclude that an individual with a low level of initial ability (i.e., competence/skill/knowledge) who must exert a high level of task-elicited effort to successfully complete a task should end up with the same level of current ability as an individual with a high level of initial ability who need only exert a low level of task-elicited effort (see Figure 4).

As an example of this, compare a student taking a class on educational psychology with no prior background in psychology (i.e., with low initial competence) to a student in the same class who has already taken an introductory psychology course (i.e., with high initial competence). Even if these students are assumed to have the same intellectual capacity, we might expect the first student to have to spend more time and effort studying for the first exam than the second student in order to earn a high grade. However, if after exerting different levels of task-elicited effort, both students receive an "A" on the exam, we may conclude that the two students have equal levels of competence in educational psychology. Considering that task-elicited effort led participants to perceive an inverse relation between effort and ability in all of our studies (Muenks et al., 2016), this type of situation (i.e., where students perceive effort to be task-elicited and maintain a

competence view of ability) might be relatively rare. Perhaps this is because a task-elicited perception of effort may, in certain cases, activate a capacity view of ability and/or an entity theory of intelligence (see Heyman & Compton, 2006, Study 3).

When students perceive effort to be task-elicited, but instead assume a capacity view of ability, their conclusions may depend in part on whether they hold an entity or incremental theory of intelligence. If they are assuming a capacity view of ability and hold an entity theory, then they should always conclude that high levels of task-elicited effort indicate low levels of current ability (as initial levels of capacity should remain constant and be identical to current levels). If, however, they hold an incremental theory, then they may conclude that high levels of task-elicited effort lead to high levels of current ability over long periods of time (see Figure 2). This, however, raises the question of whether repeatedly engaging in challenging tasks that require high levels of task-elicited effort is viewed more broadly as an act of self-initiated effort.

At this point, it is somewhat difficult to map self-initiated effort onto our framework. This is because it remains to be established whether students generally interpret self-initiated effort as something that only highly motivated and capable individuals exert, or as something that serves to increase individuals' abilities (i.e., whether ability causes effort or vice versa). If students adopt the former view, then it is possible for them to perceive a positive relation between ability and effort regardless of whether they view ability in terms of competence or capacity, and regardless of whether they hold an entity or incremental theory. However, if they adopt the latter interpretation, then it is only possible for them to perceive a positive relation if they assume a competence view of ability or an incremental view of intelligence.

In conclusion, both Nicholls (1978, 1984) and Dweck (1999) focused on how students' conceptions of *ability* determine the way they view the relation between levels of effort and ability, although Nicholls (1978, 1984) focused on developmental and situational predictors of ability conceptions, while Dweck (1999; Dweck & Master, 2009) focused on stable individual differences in ability conceptions. Based on the studies described here (Heyman & Compton, 2006; Heyman et al., 2003; Muenks et al., 2016), we suggest that perceptions of *effort* may also contribute to students' thinking about the relation between levels of effort and ability and that this contribution may be partly independent from the contribution of their perceptions of ability.

Discussion

An important determinant of students' academic motivation is their assessments of how much ability they have in various domains (e.g., math, language arts, music, etc.). Because these assessments are often based on how much effort students have expended completing tasks in those domains, it is important to identify factors that determine whether they are likely to think in terms of a positive or inverse relation between levels of effort and ability. With this aim in mind, we have reviewed the literatures pertaining to three types of factors: developmental, contextual, and individual differences. Developmentally, children appear to have trouble differentiating between concepts of effort and ability and are unable to conceive of them as inversely related to one another until they reach a certain

level of cognitively maturity. With respect to context and individual differences, certain cues (e.g., social comparison and competition) and beliefs (e.g., an entity theory of intelligence) lead students to think in terms of an inverse relation between effort and ability, whereas other cues (e.g., a classroom-wide "press for understanding"; see Middleton & Midgley, 2002) and beliefs (e.g., an incremental theory of intelligence) may lead students to think in terms of a positive relation.

Another aim of the article was to extend Nicholls's framework for understanding students' thinking about the relation between effort and ability. Specifically, our extended framework helps answer some unresolved questions in the literature, such as how an individual can quickly shift between thinking of effort and ability as being inversely versus positively related. We hope that this extended framework will help inspire researchers to design future studies that test more precise hypotheses about how students think about the relation between effort and ability across a range of different achievement situations or contexts. Findings from such studies can help educators and parents better understand the ways in which students think about effort and ability, as well as pave the way for the design and assessment of new interventions that promote students' academic motivation and engagement.

Suggestions for Future Research

Many of the studies reviewed in this article, especially those employing a vignette methodology, were published in the 1970s and 1980s. More recently, individual difference (e.g., Dweck, 1999) and goal orientation (e.g., Midgley et al., 1998) perspectives have dominated, and there has been less work directly examining developmental or contextual factors that influence the perceived relation between effort and ability. However, some newer studies (e.g., Heyman & Compton, 2006; Heyman et al., 2003; Muenks et al., 2016) have helped clarify the influence of these factors on students' thinking about effort and ability. Nevertheless, there remain some gaps in the literature and unanswered questions that future research should address.

First, researchers should test the framework represented in Figures 1, 2, and 4 in order to better understand the circumstances that lead students to shift between inverse and positive perceived relations. This would entail measuring or manipulating contextual and individual different factors within the same study. For example, researchers could manipulate students' exposure to ego-involved or task-involved context cues *and*, as part of the same study, measure students' implicit theories about intellectual ability. Hierarchical linear modeling models could also help here by providing a powerful means of simultaneously examining within-participant contextual effects and between-participant individual effects. Researchers should also begin to assess novel factors that are posited by our framework, such as the timeframe associated with effort expenditure. For example, researchers could examine whether evaluating ability based on effort exerted over short periods of time (e.g., with reference to a specific task) leads to different conclusions than evaluating ability based on effort exerted over longer periods of time (e.g., the course of semester or year); they should also examine whether this difference depends in part on the implicit theory of intellectual ability held by participants (i.e., entity vs. incremental).

Second, in the future, researchers could examine how the effects of individual difference and contextual factors on children's thinking about effort and ability change over the course of development. For example, classroom context cues may be more important or salient up to a certain age, at which point individual difference factors may begin to take precedence. Longitudinal or cross-sectional studies could help determine the specific developmental points at which various factors have the strongest influence on students' thinking.

Third, in future studies, researchers should investigate how students' perceptions of effort source might influence their thinking about the effort–ability relation. Although some researchers have begun to directly examine this (Muenks et al., 2016), more studies are needed on how students' perceptions of effort source interact with their perceptions of ability. For example, the influence of students' perceptions of effort source (i.e., whether effort is task-elicited or self-initiated) on their thinking about the effort–ability relation above may depend in part on whether they have an entity or incremental theory of intellectual ability or whether they have been primed to think about ability in terms of competence or capacity. As previously mentioned, it is also possible that task-elicited effort cues could lead participants to think about ability as a capacity or to hold an entity theory, whereas self-initiated cues could lead them to think about ability in terms of competence or to hold an incremental theory (see Heyman & Compton, 2006, Study 3). Additionally, researchers could examine whether perceptions of effort source can be reliably measured as a stable individual difference variable in addition to being temporarily influenced by environmental cues.

Fourth, researchers could examine under what conditions it may be motivationally beneficial to think about effort and ability as either positively or inversely related. Although much of the current work suggests that conceiving of effort and ability as positively related leads to better motivation and achievement (Blackwell et al., 2007; Dweck, 1999; Hong et al., 1999), it may be useful to consider whether there are any circumstances in which it is beneficial to think about effort and ability as inversely related (see "Implications" section below).

Finally, once researchers have a better sense of which factors shape students' thinking about effort and ability, as well as which ways of thinking are most beneficial in particular contexts, the next step would be to design interventions that target these factors. For example, if research on effort source determines that students' perceptions of self-initiated effort lead them to think about effort and ability as positively related, and this way of thinking is shown to produce motivational benefits in the context of interest, then future researchers could design and test interventions that target students' perceptions of effort source in this context. Ultimately, these interventions could help students regulate their own thinking about ability and effort in a manner that leads to better motivation and learning in school (see Miele & Scholer, 2016).

Implications for Practice

Although this theoretical review is meant to stimulate new research on achievement motivation that will ultimately have practical implications for teachers, parents, and school administrators, we caution that more empirical support for our proposals is needed before firm recommendations for practice can be made. That

being said, there are some important findings in the existing literature (which we reviewed in the first half of the article) that we feel can be useful for practitioners. For instance, there is much support for the idea that it is generally beneficial for students to think about effort and ability as being positively related in the long term, and perhaps in the short term as well (i.e., the benefits of holding a "growth mind-set"; see Dweck & Master, 2009; Yeager & Dweck, 2012). Researchers have created interventions that promote growth mindsets in students over extended periods of time (e.g., Blackwell et al., 2007; Paunesku et al., 2015) and have identified parent and teacher behaviors, such as effort or process praise, that communicate growth mind-sets to children (e.g., Gunderson et al., 2013; Mueller & Dweck, 1998; Pomerantz & Kempner, 2013). Additionally, much research has shown that creating a "mastery-oriented" classroom environment—which is very similar to Nicholls's (1984) definition of a "task-involved" situation—promotes beneficial aspects of students' motivation in the classroom (e.g., Ames, 1992; Park, Gunderson, Tsukayama, Levine, & Beilock, 2016).

We hope that our proposed framework will encourage research that uncovers additional nuances in this work. For example, when focusing on different levels of analysis and points in time, we may find that fostering a positive relation is not always better. For instance, consider a student who is deciding whether to major in computer science or journalism. It may be helpful for this student to consider (in addition to other factors, such as her interests) the current rate at which she acquires knowledge and skills in each domain relative to her peers (i.e., her current capacity). This may give her a sense of how hard she needs to work to keep up with her peers in each domain (at least in the beginning). It may also be helpful for this student to consider the extent to which her capacities can be increased through sustained effort over extended periods of time and the rate at which such an increase is likely to occur. This may give her a sense of how much long-term success she is likely to achieve in each domain. If this student has dyscalculia, she may find that her capacity for math increases more slowly than her capacity for verbal processing and, as a consequence, choose to major in a field, such as journalism, which she believes provides her with a greater chance of being successful.

Once students choose a domain to focus on, however, it is likely that thinking in terms of a positive relation between effort and ability yields the greatest motivational benefits. For instance, if the student with dyscalculia instead chooses to major in computer science because there are more jobs available in this sector (or because she finds this field to be particularly interesting), it may be helpful for her to focus on the extent to which her efforts lead to increases in competence (in the short term) and capacity (in the long term) as opposed to possible innate limits of this capacity. Thus, although in the long term, it seems universally beneficial to think about a positive relation between effort and ability, it might be helpful, for various reasons, to think about an inverse relation in the short term. However, more research is needed to empirically test these claims.

Notes

Both authors contributed equally to the article.

[1]It is worth noting that this component of Nicholls's (1984) theory is conceptually similar to contemporary goal orientation theory (see Kaplan & Maehr, 2007, for a review).

[2]It is worth noting some key differences between Nicholls's (1984) and Dweck's (1999; Dweck & Master, 2009) use of terminology. Nicholls made a conceptual distinction between "ability" and "intelligence," such that the term "intelligence" naturally presumes a conception of ability as a capacity that is differentiated from effort, while "ability" refers to something that can be conceptualized in multiple ways (i.e., as differentiated or undifferentiated from effort; see Nicholls, Patashnick, & Mettetal, 1986). Dweck, however, used the terms "intelligence" and "ability" somewhat interchangeably (e.g., Dweck, 2002); thus, for the purposes of this review article, we will discuss Dweck's (1999; Dweck & Leggett, 1988; Dweck & Master, 2009) theory using the term *ability*, rather than intelligence, to be consistent with Nicholls's terminology and because Dweck used the term frequently when discussing her own theory. However, it is important to note that concerns have been expressed in the literature about whether or not the two terms meaningfully differ from one another.

[3]In their article, Heyman and Compton referred to the "intentional effort" condition as simply the "effort" condition. We have relabeled it here to more clearly distinguish between the actual effort expended by the characters (in terms of time) and the framing of effort in terms of "trying" or "difficulty." The study also included a *no prime* condition.

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